

Simple Linear Regression – Marketing ROI Analysis

Importing Libraries

```
# Step-by-Step: Simple Linear Regression - Marketing ROI Analysis
# Importing necessary libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
from statsmodels.stats.diagnostic import het_breuschpagan
from scipy import stats
```

Setting style for plots

```
# Setting style for plots
plt.style.use('seaborn-v0_8')
sns.set_palette("husl")
%matplotlib inline
```

Uploading data-set

```
# Uploading data-set
import ipywidgets as widgets
from IPython.display import display

uploader = widgets.FileUpload(accept='.csv,.xlsx', multiple=False)
display(uploader)

FileUpload(value=(), accept='.csv,.xlsx', description='Upload')
```

Checking if data is loaded successfully

```
# Checking if data is loaded successfully
print(df.head())
print(df.shape)
```

	TV	Radio	Social_Media	Sales
0	16.0	6.566231	2.907983	54.732757
1	13.0	9.237765	2.409567	46.677897
2	41.0	15.886446	2.913410	150.177829
3	83.0	30.020028	6.922304	298.246340
4	15.0	8.437408	1.405998	56.594181

(4572, 4)

Exploring data-set

```
# Exploring data-set
print("Dataset Shape:", df.shape)
print("\nFirst 5 rows:\n", df.head())
print("\nData Info:")
df.info()
print("\nSummary Statistics:\n", df.describe())
```

Dataset Shape: (4572, 4)

First 5 rows:

	TV	Radio	Social_Media	Sales
0	16.0	6.566231	2.907983	54.732757
1	13.0	9.237765	2.409567	46.677897
2	41.0	15.886446	2.913410	150.177829
3	83.0	30.020028	6.922304	298.246340
4	15.0	8.437408	1.405998	56.594181

```
Data Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4572 entries, 0 to 4571
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0    TV          4572 non-null   float64
1    Radio        4572 non-null   float64
2    Social_Media 4572 non-null   float64
3    Sales        4572 non-null   float64
dtypes: float64(4)
memory usage: 143.0 KB

Summary Statistics:
```

	TV	Radio	Social_Media	Sales
count	4572.000000	4572.000000	4572.000000	4572.000000
mean	54.066857	18.160356	3.323956	192.466602
std	26.096461	9.672723	2.211218	93.071947
min	10.000000	0.000684	0.000031	31.199409
25%	32.000000	10.546519	1.529879	112.397660
50%	53.000000	17.875181	3.057150	189.632530
75%	77.000000	25.647476	4.805832	272.414914
max	100.000000	48.871161	13.981662	364.079751

✓ Checking for missing value

```
# Checking for missing values
print("\nMissing Values:\n", df.isnull().sum())
```

```
Missing Values:
TV          0
Radio       0
Social_Media 0
Sales       0
dtype: int64
```

✓ Handle missing values

```
# Handle missing values
# Fill with mean (for numerical columns)
if df.isnull().sum().any():
    df = df.dropna()

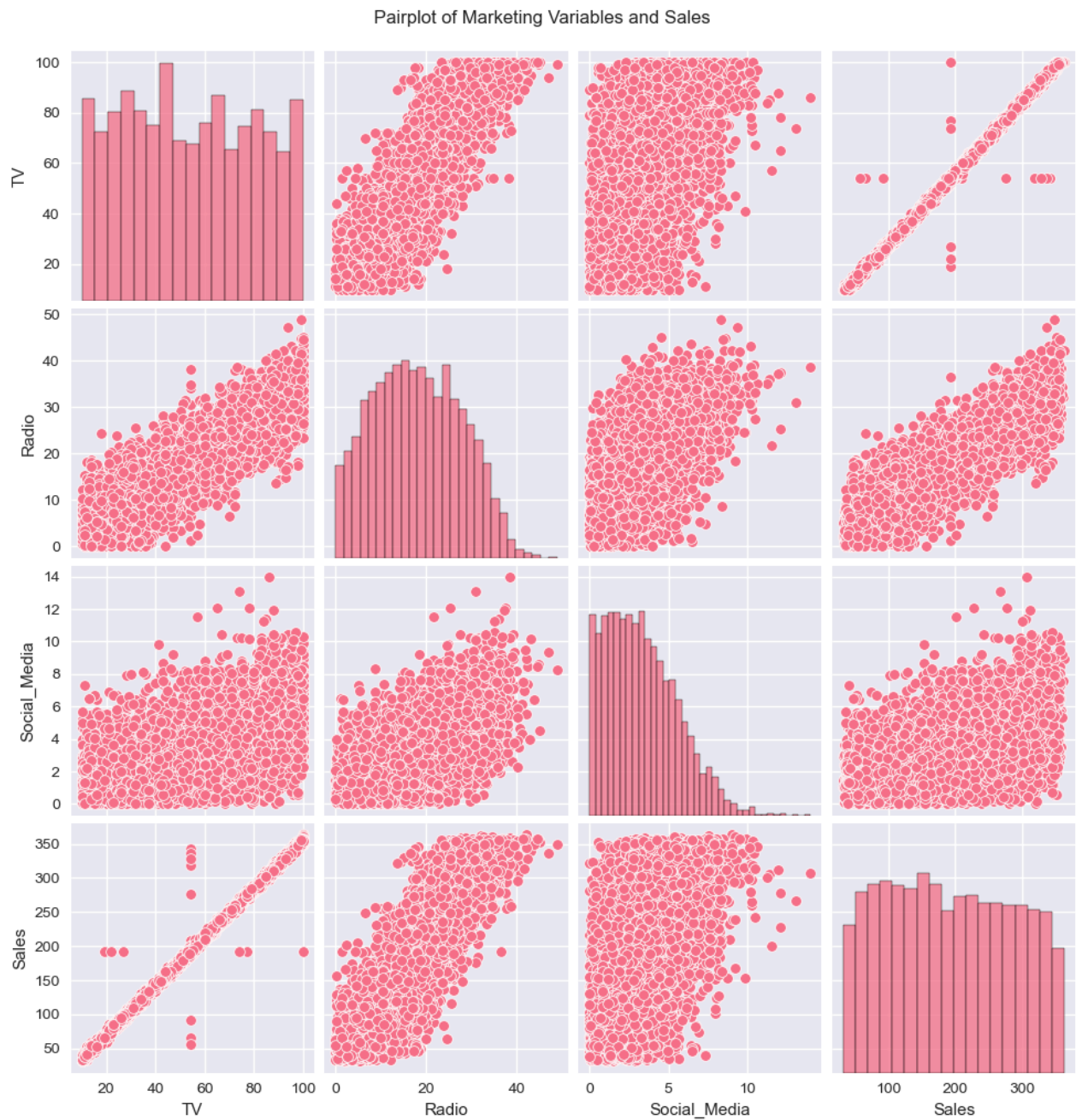
print("\nAfter handling missing values - Shape:", df.shape)
```

```
After handling missing values - Shape: (4572, 4)
```

✓ Performing Exploratory Data Analysis (EDA) with Visualizations

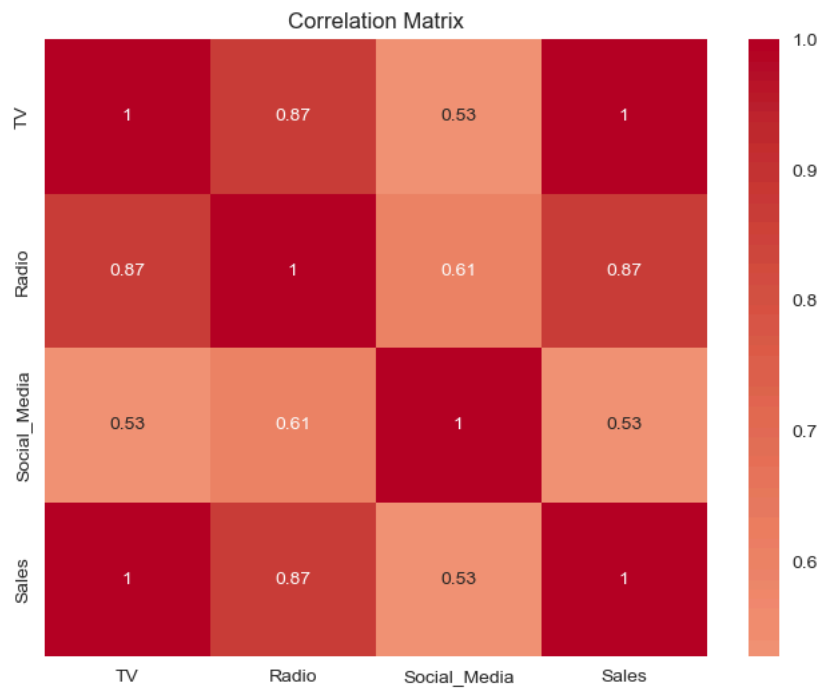
✓ Pairplot to see relationships

```
# Pairplot to see relationships
sns.pairplot(df)
plt.suptitle('Pairplot of Marketing Variables and Sales', y=1.02)
plt.show()
```



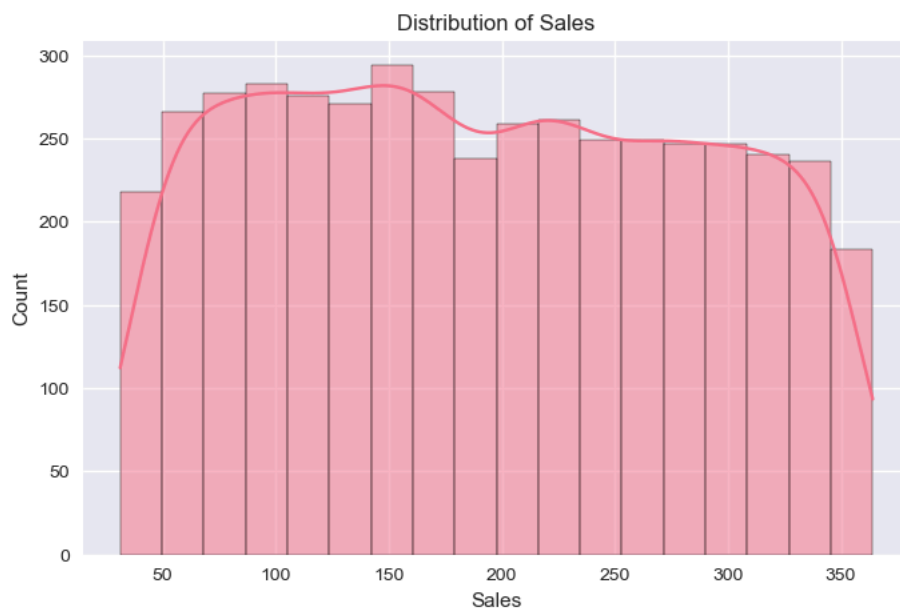
Correlation heatmap

```
# Correlation heatmap
plt.figure(figsize=(8, 6))
correlation_matrix = df.corr()
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', center=0)
plt.title('Correlation Matrix')
plt.show()
```



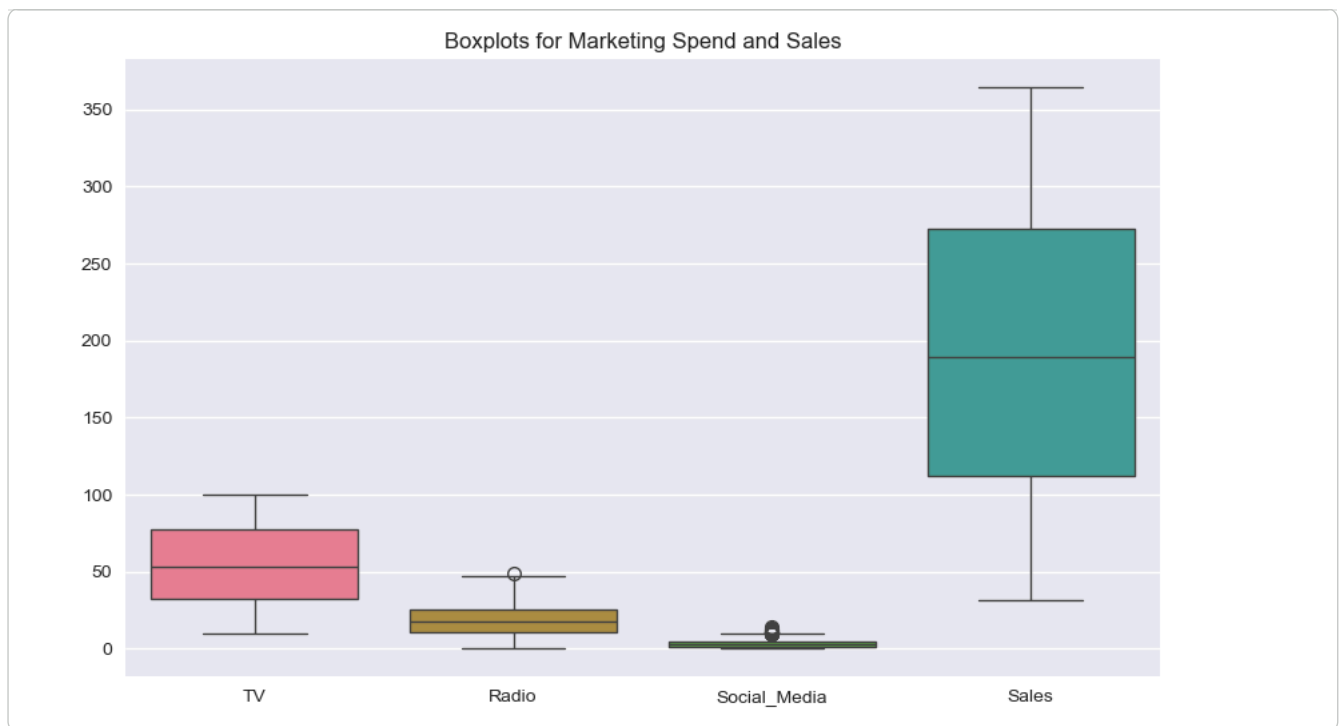
✓ Distribution of Sales (target variable)

```
# Distribution of Sales (target variable)
plt.figure(figsize=(8, 5))
sns.histplot(df['Sales'], kde=True)
plt.title('Distribution of Sales')
plt.show()
```



✓ Boxplots for outliers

```
# Boxplots for outliers
plt.figure(figsize=(10, 6))
sns.boxplot(data=df[['TV', 'Radio', 'Social_Media', 'Sales']])
plt.title('Boxplots for Marketing Spend and Sales')
plt.show()
```



Identifying the Independent Variable Most Correlated with Sales

```
# Identifying the Independent Variable Most Correlated with Sales
independent_vars = ['TV', 'Radio', 'Social_Media']
```

```
correlations = {}
for var in independent_vars:
    if var in df.columns:
        corr = df[var].corr(df['Sales'])
        correlations[var] = corr
        print(f"Correlation between {var} and Sales: {corr:.4f}")
```

```
Correlation between TV and Sales: 0.9967
Correlation between Radio and Sales: 0.8674
Correlation between Social_Media and Sales: 0.5281
```

Finding the strongest predictor

```
# Finding the strongest predictor
best_var = max(correlations, key=correlations.get)
print(f"\n✅ Strongest predictor: {best_var} (correlation: {correlations[best_var]:.4f})")
```

```
✅ Strongest predictor: TV (correlation: 0.9967)
```

Simple Linear Regression with the best independent variable

```
# Simple Linear Regression with the best independent variable
X = df[best_var]      # Independent variable
y = df['Sales']       # Dependent variable
```

```
# Adding constant for intercept
X = sm.add_constant(X)
```

```
# Fitting the model
model = sm.OLS(y, X).fit()
```

```
# Printing summary
print(model.summary())
```

```

                        OLS Regression Results
=====
Dep. Variable:          Sales    R-squared:                0.993
Model:                  OLS      Adj. R-squared:           0.993
Method:                 Least Squares   F-statistic:          6.792e+05
Date:                   Tue, 09 Jun 2026   Prob (F-statistic):    0.00
Time:                   04:01:58   Log-Likelihood:       -15765.
No. Observations:       4572      AIC:                  3.153e+04
```

```

Df Residuals:      4570    BIC:      3.155e+04
Df Model:          1
Covariance Type:  nonrobust
=====
              coef    std err          t      P>|t|      [0.025    0.975]
-----
const         0.2849      0.259      1.100      0.271     -0.223     0.793
TV            3.5545      0.004     824.111      0.000      3.546     3.563
=====
Omnibus:            2844.142    Durbin-Watson:      2.009
Prob(Omnibus):      0.000    Jarque-Bera (JB):    12214106.398
Skew:              1.223    Prob(JB):          0.00
Kurtosis:          256.200    Cond. No.         138.
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

✓ Creating a diagnostic plot

✓ 1. Residuals vs Fitted (Linearity & Homoscedasticity)

```

# 1. Residuals vs Fitted (Linearity & Homoscedasticity)
fitted = model.fittedvalues
residuals = model.resid

plt.figure(figsize=(10, 6))
sns.scatterplot(x=fitted, y=residuals)
plt.axhline(y=0, color='r', linestyle='--')
plt.xlabel('Fitted Values')
plt.ylabel('Residuals')
plt.title('Residuals vs Fitted Values')
plt.show()

```

[Show hidden output](#)

✓ 2. Q-Q Plot (Normality of Residuals)

```

# 2. Q-Q Plot (Normality of Residuals)
plt.figure(figsize=(8, 6))
sm.qqplot(residuals, line='s')
plt.title('Q-Q Plot of Residuals')
plt.show()

```

[Show hidden output](#)

✓ Scale-Location Plot (Homoscedasticity)

```

# 3. Scale-Location Plot (Homoscedasticity)
plt.figure(figsize=(10, 6))
sns.scatterplot(x=fitted, y=np.sqrt(np.abs(residuals)))
plt.axhline(y=0, color='r', linestyle='--')
plt.xlabel('Fitted Values')
plt.ylabel('√|Residuals|')
plt.title('Scale-Location Plot')
plt.show()

```

[Show hidden output](#)

✓ Breusch-Pagan test for heteroscedasticity

```

# Breusch-Pagan test for heteroscedasticity
bp_test = het_breuschpagan(residuals, model.model.exog)
print(f"Breusch-Pagan test p-value: {bp_test[1]:.4f} (p > 0.05 suggests homoscedasticity)")

```

Breusch-Pagan test p-value: 0.8720 (p > 0.05 suggests homoscedasticity)

✓ Interpreting the result

✓ Extract key metrics

```
# Extract key metrics
r_squared = model.rsquared
coef = model.params[best_var]
p_value = model.pvalues[best_var]
intercept = model.params['const']

print(f"""
=== Business Interpretation ===

R-squared: {r_squared:.4f}
→ This means that {r_squared*100:.1f}% of the variation in Sales can be explained by spending on {best_var}.

Coefficient for {best_var}: {coef:.4f}
→ For every additional $1 spent on {best_var}, Sales increase by approximately ${coef:.2f} on average (holding other factor

p-value: {p_value:.4f}
→ Since p-value < 0.05, the relationship is statistically significant.

Intercept: {intercept:.2f}
→ Baseline Sales when {best_var} spend = $0.
""")
```

```
=== Business Interpretation ===

R-squared: 0.9933
→ This means that 99.3% of the variation in Sales can be explained by spending on TV.

Coefficient for TV: 3.5545
→ For every additional $1 spent on TV, Sales increase by approximately $3.55 on average (holding other factors constant).

p-value: 0.0000
→ Since p-value < 0.05, the relationship is statistically significant.

Intercept: 0.28
→ Baseline Sales when TV spend = $0.
```

▼ ROI Calculation (simple)

```
# ROI Calculation (simple)
```